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Do global food systems have an Achilles heel? The potential for regional food systems to support resilience in regional disasters

Rebekah Paci-Green[a] and Gigi Berardi[b]

[a]Assistant Professor; Environmental Studies, Western Washington University, 516 High Street, Bellingham WA, 98225; Rebekah.paci-green@wwu.edu
[b]Professor; Environmental Studies, Western Washington University, 516 High Street, Bellingham WA, 98225; Gigi.berardi@wwu.edu

Corresponding author: Rebekah Paci-Green, Rebekah.paci-green@wwu.edu, 360-650-2707

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Abstract

Today's domestic United States food production is the result of an industry optimized for competitive, high yielding, and high-growth production for a globalized market. Yet, industry growth may weaken food system resilience to abrupt disruptions by reducing the diversity of food supply sources. In this paper, we first explore shifts in food consumption towards reliance upon complex and long-distance food distribution, food imports, and out-of-home eating. Second, we discuss how large-scale, rapid-onset hazards may affect food access for both food secure and insecure households. We then consider whether and how regional food production might support regional food resilience. To illustrate these issues, we examine the case of western Washington, a region rich in agricultural production, but also threatened by a Cascadia Subduction Zone earthquake and tsunami. Such an event is expected to disrupt transportation and energy systems on which the dominant food distribution system relies. Whether a regional food supply — for the purposes of this paper, defined as food production in one or adjacent watersheds — can support a broader goal of community food resilience during large-scale disruption is a key theme of our paper. The discussion that ensues is not meant to offer simplistic, localist solutions as the one answer to disaster food provision, but neither should regional food sources be dismissed in food planning processes. Our exploration of regional farm production, small in scale and flexible, suggests regional production can help support food security prior to the arrival of emergency relief and retail restocking. Yet in order to do so, we need to have in place a robust and regionally appropriate food resilience strategy. This strategy should address caloric need, storage, and distribution, and, in so doing, re-balance our dependence on food supplied through imports and complex, domestic supply chains. Clearly, diversity of food sourcing can add redundancy and flexibility, allowing more nimble food system adaptation in the face of disruption.

Keywords: food access; disasters; resilience; emergency relief; earthquake; Washington; small-grain cereals; regional food systems

Introduction

On any given day in the United States (U.S.), food appears to be plentiful, even overly abundant. This is the land of box store grocery market and discount food clubs that sell instant noodles in packs of two dozen at a time. It is also the land of fast food restaurants, with more fast food restaurants per resident than any other country in the world — 225 per million in 2015 according to *The Economist*. To supply it all, vegetables from California's bountiful Central Valley, staple crops of corn and soy from the Midwest, and dairy products from the northern and western states crisscross the nation. At the same time, a growing supply of imported food augments domestic production. While this abundance is far from equally distributed — swaths of the southeast and Appalachia have been highlighted as food deserts by the United States Department of Agriculture (USDA) — for most communities an abundance and a plethora of food choice is generally a given.

Yet, not all days or years are the same. On average, between the years 1960 and 2013, the U.S. experienced $12.1 billion in property losses and $2.8 billion in crop losses annually from extreme natural processes such as hurricanes, flooding, severe weather, landslides, tornados, and earthquakes (HVRI 2013). These geological and hydro-meteorological processes communities experience occur within historical, socio-political, and economic contexts that often exacerbate the natural processes and create the infrastructure fragility and societal vulnerabilities that lead to heterogeneous geographic and temporal patterns of loss we commonly call 'disaster.' As such, disasters, and even the magnitude and frequency of some hazards, are never truly 'natural.' They are the product of a dynamic environment and human agency (Oliver-Smith 1999; Steinberg 2000; Wisner et al. 2003). Compounding the effect of the hazards is infrastructure fragility and uneven social vulnerability, such that local and regional transportation, energy systems, and food distribution networks in affected communities are often compromised.

The USDA defines food security as access by all people at all times to enough food for an active, healthy life (Coleman-Jensen, Gregory, and Singh 2014). Food security thus can be considered as access to a daily minimum amount of culturally appropriate calories. Food security is threatened when:

a) Civil conflict or limited access to food-producing resources, especially land, result in food deficits,
b) Natural hazards strike vulnerable populations, whether that be a broad segment of a community or, more frequently, its marginalized members, or
c) Food quality is compromised, due to either a) or b) above, or to inadequate quality of food (little fresh produce, poor protein sources, low-nutrient calories) and/or reasonably priced retail outlets selling high quality foods.

Little research in the U.S. has been conducted on changes in food access and food security during the immediate emergency phase of disasters. Multi-year drought can radically reduce crop production in a region. Large magnitude floods and hurricanes may damage storage facilities and distribution networks. Floodwaters, mixed with industrial waste and sewage, can directly destroy or contaminate food resources; debris and silt from floodwaters can compact soil and leave fields unusable for years. With floods, however, communities often have sufficient warning regarding rising rivers and predicted hurricane paths so that many households can evacuate to places outside the affected region, or stockpile foods out of harm's way.[1]

For rapid-onset events such as earthquakes and some landslides, which occur with little or no warning, damage to food supply and distribution can be more unexpected. It can directly impact food access for households at all income

[1] Regardless of type of disaster, evacuation and stockpiling are premised on sufficient resources. For low-income households, stockpiling and evacuation may be unaffordable. Households without private transportation must rely upon social networks or already overwhelmed public and emergency transport to evacuate. Many often choose to ride out the events at home, as was seen in New Orleans during Hurricane Katrina.

levels; households simply do not have time to stockpile or evacuate after they realize the event is occurring. News reports from Hurricanes Katrina and Sandy in the U.S., reports from the 2010 New Zealand Christchurch earthquake, and limited research after the 2011 Tohoku earthquake and tsunami in Japan, offer some limited insights into how large, rapid-onset hazards may affect food access and purchasing behavior in the U.S.[2]

Following the Tohoku Earthquake, many residents experienced psychological anxiety, especially around the availability of food. A week after it struck, the Minister of State for Consumer Affairs pleaded with citizens to avoid hoarding. In the Tokyo metropolitan area, over 300 kilometers from the epicenter, Hori and Iwamoto (2014) found that consumers engaged in panic buying for a wide range of products, although actual shortages occurred only for a few items. Supply in the metropolitan area was robust, nevertheless panic buyers emptied shelves, triggering stress and panic purchasing in others. While empty shelves in Tokyo did not create actual food insecurity, the picture was different near the epicenter. In districts of East Japan, where damage to roads and power grids was pronounced, households faced serious supply shortages for many items in the chaotic weeks after the event.[3] At evacuation centers, survivors subsisted on rice balls and bread for many weeks, in part due to fears of radiation exposure through contaminated foods grown near the meltdown of the Fukushima-Daiichi nuclear plant (Belyakov 2015).

Following Hurricane Katrina in 2005, New Orleans and other Gulf Coast residents were similarly stranded by washed away roads, floodwaters, and fallen power lines and trees. While meteorologists did provide early warning for this event, thousands of low-income, elderly, and other New Orleans residents remained in the city when the levees failed and rapid-onset flooding began. With residents trapped in the city, both water and food supplies dwindled. Supply convoys headed to New Orleans took three days to first reach the city, at which point distribution to those in need was still necessary; specialized supplies took longer (Thomas and Mora 2014). Even after the emergency response phase had subsided, impacts on food access lingered. The number of supermarkets in New Orleans dropped from 36 stores to 15 stores. Residents without transportation relied more on local convenience stores, resulting in a drop in household consumption of fresh food (Mioses 2007). A multi-year study of supermarket access found that racial-ethnic disparities in access to supermarkets increased after the storm and did not return to pre-event levels for four years (Mundorf et al. 2015). Years later and two thousand kilometers north, "Superstorm Sandy" slammed into the northeast U.S. in 2012. Even with extensive early warning, supermarkets in coastal areas of New Jersey and New York threw out food supplies contaminated by the meter or more of storm surge that many experienced (Taylor 2012). In the immediate aftermath of these events, retail food availability shrank and access to specific items was unpredictable.

The Christchurch earthquake offers another glimpse of how large-scale, rapid-onset hazards can reduce food security in urban centers. Following the earthquake, the retail outlets, warehouses, and stocks of commercial food retailers were damaged. Residents of Christchurch could not purchase basic foods such as milk and bread and while supplies were brought in from other regions, the damage to transportation routes, including road, rail, and port facilities, slowed their delivery. Access to water also complicated food security. Eighty percent of the water system was severely damaged. For two months, many residents had no running water and relied upon water brought by tanker to centralized locations, a system that was not available in the immediate days and weeks after the quake (McColl and Burkle 2012). Computational modeling for Los Angeles provides further nuance and suggests that degradation of water supply has broader impacts than simply food security. A computable general equilibrium analysis of earthquake-induced damage to the California Aqueduct found that without aggressive conservation

[2] The World Health Organization and other United Nations agencies routinely complete rapid food security assessments of endemic and disaster-induced food insecurity in low-income countries. While crucial for humanitarian response to these events, the differences in subsistence, income, and spatial distribution of population make these assessments less useful for deducing disaster impacts to food access in the United States.
[3] A less extreme case of food shortage made international news when New Zealanders faced a shortage of Marmite spread several months after the 2011 Christchurch earthquake.

strategies, Los Angeles' economy would experience significant negative impacts due to loss of water for cleaning, cooling, and manufacturing, even for a closure as short as six months (Rose et al. 2012).

The specter of spoiled food, empty shelves, and restricted water flow out of the kitchen faucet has prompted emergency preparedness public education for decades. These campaigns typically recommend that households store at least three days of food, water, and medication, though many emergency managers privately admit that a three-day supply may not be adequate in large-scale disasters. Two weeks of supplies — more in remote areas — is what they would rather see communities stockpiling. However, it is not clear how many households in hazard-prone areas even meet the three-day stockpile recommendations. One study sheds light. Thomas and Mora (2014) used interviews of 172 householders in Wellington, New Zealand to understand household water, food, and medicine stockpiles that might be available immediately after a large-scale earthquake. They found only 35 percent of respondents reported being prepared to survive for three days, though the researchers' own calculations of supplies suggested just over half of the households had sufficient resources for the recommended three days. While respondents reported that they would be quite likely to offer assistance to others, they indicated that they would be less likely to do so after a week of shortages. In fact, they would be more likely to lie about their resources as the disaster continued. Respondents also indicated that, after one week, they would be willing to engage in novel and higher risk behavior around food and water access — taking water from unpurified sources and stealing essential resources.

Even with limited literature on the topic, the potential for large-scale, rapid-onset natural hazards to disrupt food access in the immediate impact zones is a concern. Communities at risk, which for the U.S. includes large swaths of the country, might rightly ask what can be done now to better ensure their food system remains resilient. How could access continue for all during such disruptions? We argue that part of the answer may lie in regional food systems.

Can regional food production help?

Much has been written about the "local" in food, but little in terms of food interruptions and the regional contribution to increasing food system resilience (Dubbeling 2013; Ruhf 2015). From previous research, we have found that, on a regional level, small- and medium-sized farms contribute substantially to food supply and economy, as well as model resilience (Hammond et al. 2013). Ruhf (2015) further argues that regional food systems are better drivers of resilience as they reduce dependence on external resources, such as are present in long-distance supply chains, and help develop a nimble and flexible regional capacity.

We suggest here that interest in a regional agriculture should not be only the domain of self-proclaimed foodies, locavores, and small-scale producers. Certainly, all of these groups contribute to a vibrant regional agricultural economy, but we feel that the locavore topic has been well covered in numerous other publications (Clancy and Ruhf 2010; Low and Vogel 2011), with Kloppenburg, Hendrickson, and Stevenson's (1996) "Coming in to the Foodshed," being a particularly insightful, early treatment of the subject. Yet, beyond these "true believers," people of all warp and weave have interests in adequate food access during crises, and in the pivotal role such access would have in supporting community recovery. Whether a regional food supply — for the purposes of this paper, defined as food production in one or adjacent watersheds — can support a broader goal of community food resilience during large-scale disruption becomes a question of interest for all. The discussion that ensues is not meant to offer simplistic, "localist" solutions as the one answer to disaster food provision. Likewise, dismissing local food sources is similarly unhelpful to food planning processes. Rather, diversity of food sourcing can add redundancy and flexibility, allowing more nimble food system adaptation in the face of disruption.

The purpose of this paper is threefold. First, we will explore shifts in food consumption patterns towards reliance upon complex and long-distance food distribution, food imports, and out-of-home eating. At the same time, food insecure households increasingly rely on food pantries, themselves supported by federal supplies trucked into a

region in a complex food distribution system. Second, we discuss how large-scale, rapid-onset natural hazards may affect food access for both food secure and insecure households, using the case of a future subduction-zone earthquake in Washington state as a point of discussion. Population considerations, infrastructure impacts, implications for food distribution, immediate food access issues, and estimates of population needs in the disaster event are discussed. Third, we consider whether and how regional food production, especially small grains — and, to a certain extent, beans, and nuts — might support regional food resilience, at least in terms of calories, during disasters. We also consider challenges to regional storage and distribution.

Clearly, regional production and distribution should be considered prominently in national resilience planning, as an antidote to the actions of Federal agencies with an "almost incomprehensible web of sub-agencies whose ability to function in an actual crisis is questionable" (Endres and Endres 2009, p 407). Regional food production can support efforts at other scales — infrastructure mitigation at the state level and preparedness at the household level — to reduce the likelihood that households experience food insecurity in the 'gap' between when a rapid-onset hazard occurs and emergency relief has reached their communities. However, we argue that regional food production systems are a limited resource, if not combined with adequate regional storage and distribution strategies.

This article consists primarily of a review of published work on food imports, community food supply strategies, diet and consumption patterns, food industry and agricultural production data, and farm-to-food information. Initial research for this paper was conducted under a small Federal Emergency Management Agency grant for the purpose of analyzing food system resilience; we have expanded on that research here. We look at immediate and short-term food deficits resulting from quick-onset natural hazards, rather than ubiquitous and persistent vulnerabilities, which are the rightful concern of a welfare state, and the subject of important discussion elsewhere (Poppendieck 1994).

As Endres and Endres' (2009) suggest in their provocative history of U.S. food security, resilience planning is very much a function of regularly-occurring threats (much like disastrous airline incidents prompt greater safety measures). At the same time, Fraser et al. (2015) argue that state-sponsored strategies of strategic grain stores in the past and especially for the poor, have declined in favor of market-oriented approaches that cloud assessments of food system resilience. In rapid-onset hazard events, a market-oriented approach may result in food insecurity for some households, an experience that can increase anxiety, risky behavior, long-term trauma, or lead to regional outmigration. For highly vulnerable individuals, such food insecurity may prove to be fatal.

We maintain that consideration of rapid-onset hazards and their impacts is a useful exercise for assessing food system resilience and that disaster scenarios can serve to push us in planning for increased resilience. An intriguing question is: Could extreme situations prompt communities to consider a greater regional reliance in more ordinary times? While we cannot provide definitive answers, the question remains an underlying theme of this work.

Declining resilience: changing patterns in food access

Domestic U.S. food production is the result of a competitive, growth-oriented industry. Most current growth comes from international market expansion, which includes the purchasing of entire farm operations, increasing contract crop production, and emerging proprietary companies operating abroad (IRS 2014). Increases in a company's market share usually have come at the expense of a competitor's loss of share, which has led to the consolidation of smaller farms and companies into larger conglomerates with high degrees of differentiation and centralized production and warehousing. According to the Internal Revenue Service, such concentration takes advantage of federal tax incentives (Berardi et al. 2011; IRS 2014). Nonetheless, this growth weakens community resilience by ultimately reducing the diversity and scale of food supply sources. This happens as smaller operations face heavy competition in pricing and declining profit margins against large firms.

In a globalized economy, regional agricultures are influenced by market forces as much as by factors of physical geography — e.g., climate and precipitation: wheat is typically grown under 20 inches of precipitation or less in the Great Plains, cotton and tobacco in the warmer areas of the southeast, dairy is found in cooler, northern regimes. An interstate highway system, preceded by the advent of refrigerated transport by rail, opened up production and marketing channels throughout the U.S.; food moves easily from one place to another. The gradient is more economic than physical; food moves and is marketed as befits business plans of all scales.

It can be argued that food system resilience is decreasing in the U.S. in a number of different ways. At the consumer level, what and where households purchase food is shifting dramatically. Increasingly, households are moving away from relying on home-prepared meals to either eat out-of-home meals or more prepared and packaged meals in-home (Guthrie et al. 2013). As such, fewer households have adequate food (shelf-stable prepared foods and certainly less of cereals, beans, and nuts as staples) for use in disasters and during an unexpected interruption in food availability at retail grocers, discount warehouses, convenience stores, or restaurants. For households experiencing food insecurity, centralized food distribution through federal benefits and food banks has also unwittingly decreased resilience during disasters by being heavily reliant upon distance food sources and complex supply chains.

Household food consumption

In 2013, the USDA's Economic Research Services (ERS) reported total U.S. household spending on food at 1.4 trillion dollars. Half of U.S. household spending was on retail food purchases such as in restaurants (see Table 1). The other 50 percent of household spending was on groceries for home consumption from a variety of sources. U.S. households used only a tiny fraction of their food budget to purchase food at farmers' markets, through Community Supported Agriculture (purchased farm) shares, or to make direct farm purchases. Consequently, the reliance upon out-of-home meals has grown precipitously in the past decades. Expenditure on food away from home increased from 26 percent in 1960, to 33 percent in 1970, and increased to above 40 percent by 1990. With significantly fewer meals eaten at home than in earlier decades, it is safe to assume households have less staples on hand for preparing meals — including fewer canned goods and smaller stores of grains, oils, and other basic ingredients for meal preparation.

Table 1. Percentage Breakdown of U.S. Food Purchases in 2013, Total Food Purchases Equaling $1.4 Trillion (based on various datasets, USDA ERS, food expenditures)

Food Purchase Type	**%**
Food away from home	50
Food stores	31
Other stores	13
Farmers, manufactures, and wholesalers	4
Home delivery and mail order	2

The shift in food consumption has also changed what is eaten at home. Today, much of home consumption centers on prepared foods (as indicated in Table 2). Sales of convenience foods, such as prepared meals, meal replacement bars, and frozen foods, are increasing among dual-income households and consumers who are generally short on time, resulting in a shift in commodity demand (U.S. Dept. of Commerce 2008). More prepared foods invariably means a greater dependence on processed cereal grains, such as oats, wheat, and corn, as well as highly-contested (due to environmental and nutritional arguments) soy protein and oils. For many households, food security is now defined, not by the full pantry, but by the availability of nearby prepared food, whether that food is handed to them through a drive through window, in a take-out order, or in the packaged meal purchased at a nearby grocery store.

Table 2. Home Food Purchase Commodities in 2006 (based on U.S. Dept. of Commerce 2008).

Commodity	%
Prepared foods	36
Animal proteins	23
Fruits and vegetables	17
Bakery products and grains	13
Dairy products	11

Meeting the increase in the demand for prepared foods is made possible, in part, by policy derived from the North American Free Trade Agreement (NAFTA) and General Agreement on Tariffs and Trade (GATT); many domestic food companies entered into alliances with foreign entities as a result of the enactment of the afore-mentioned legislation (U.S. Dept. of Commerce 2008). Food from Canada, Mexico, China, and India dominate U.S. imports by value, with imports from Mexico and China into the U.S. experiencing over 10 percent growth between 2000 and 2014. For all categories but meat and dairy, the USDA's Economic Research Service reports an increase in volume of imports (ERS 2015). Most dramatic has been the increase in sugar and candy, with a 19.9 percent increase between 2000 and 2013. The U.S. Food and Drug Administration estimates food imports into the U.S. have increased from 4.4 million lines in 2002 to 8.6 million import lines[4] in 2010 (National Agricultural Statistical Service 2013). These food imports, coming through U.S. ports of entry, now represent over ten percent of all food consumed by U.S. households, including half of fruits and vegetables and over three quarters of seafood.

The complex distribution system these imports represent, combined with a decline in household level food storage, means food security for many today is intimately intertwined with transportation, energy, complex supply chains, and regulation systems affecting delivery of both domestic and imported products. This food system relies heavily upon a transportation infrastructure that includes ports, interstate road and rail networks, and transportation fuel systems. A regional disaster that interrupts such systems, by degrading or disrupting travel routes and ports of entry or reducing access to fuels used in food transport, poses a direct threat to food system and community resilience, both in the immediate impact zone, but also further afield. This is especially true given that produce in retail grocery stores travels about 4,200 miles, on average (Weber and Matthews 2008).

Consumption in food insecure households

Even while the commercial food distribution system providing food access for U.S. households has become increasingly complex and reliant upon transportation and energy systems, a significant percentage of households rely upon non-commercial food distribution, that is, food distributed through food banks and pantries. These sites have been the traditional means of addressing shortfalls. Yet, an examination of emergency food distribution — food distributed through food banks and pantries — in Washington state helps illuminate the fragility of these supposed safety nets.

One in six Washington households, or 14.3 percent, was food insecure during 2011-2013 (Colman-Jensen et al. 2014). In 2013, 1.3 million individuals, equivalent to a fifth of the state's population, made 8.5 million visits to food banks and pantries in Washington state. On average, each client visited a food pantry 6.5 times in 2013 (Robinson 2014). For clarity, these statistics and programs are not associated with, nor do they include USDA's Supplemental Nutrition Assistance Program (SNAP), commonly referred to as food stamps.

[4] An import "line," whether it accounts for one or one thousand items, is a data element provided by an import broker.

While annual statistics about food pantry usage suggest a large ongoing food insecurity, a regional disaster is certain to swell the ranks of households seeking access to food through food pantries. Unlike the Wellington residents studied by Thomas and Mora (2014), Washington state residents have not recently experienced a large-scale disaster that would have spurred emergency preparedness and stockpiling of adequate food, water, and medicine. Furthermore, Thomas and Mora (2014) noted that seeking food aid in a disaster, such as queuing up for a bag of groceries at the local food pantry, is more highly associated with anxiety over running out of food than a true measure of food deficit. Those more anxious about being prepared for disasters were simultaneously more likely to report increased preparedness and stated they would still seek out aid. An examination of the food pantry resources suggests this emergency food aid system may become quickly overwhelmed in a disaster.

Washington state has a highly centralized network of food assistance primarily relying on federal aid programs. Each month, food banks in Washington state receive and distribute over 554 truckloads of USDA food (Robinson 2014). USDA food, trucked directly to food banks, accounts for nearly a fifth of the food resources pantries distribute. The majority of the remaining food resources consist of food purchased or donated from retail outlets with supply chains that bring food from out of state and out of country.

Food pantries are not unlimited sources of emergency food aid. Food pantries typically stock one to three days of food supply for their customary client base. Providing adequate nutrition to populations of various vulnerabilities during a regional, rapid-onset disaster would put significant added demands upon this resource, and reduce food access especially for low-income households who regularly rely upon pantry donations (Green and Cornell, 2014).

To highlight such issues, we present a disaster scenario below. However, it is important to note that not everyone is affected equally in a disaster. Those already at risk for adequate food supply would be at a proportionately greater risk. Such increased vulnerability necessarily needs to figure into the discussion.

Food access during a rapid-onset disaster: the case of western Washington

As discussed above, United States households are increasingly reliant upon a consolidated agricultural sector and low-priced food imports, supporting a shift in consumption towards highly processed foods, food consumption outside of the home, and imported seafood and produce. These changes, combined with the reliance of food insecure households on food pantries that source significant quantities of food from outside the region, affects food insecurity during rapid-onset hazards. We explore the implications of these changes by examining a very real threat — a Cascadia Subduction Zone earthquake — to western Washingtonian's access to food during a chaotic, immediate post-event emergency.

In the event of a subduction zone earthquake, entire areas of Washington, such as the western side of the state where over five million people reside, are vulnerable to food system disruption. A 1,000-kilometer-long fault, lying mostly off shore of the Pacific coastline where the Pacific tectonic plate is subducting under the North American Plate, is capable of producing a mega-earthquake — a seismic event of magnitude 8.0 and above, which would be among the largest earthquakes ever experienced. Paleontological records indicate the last subduction zone earthquake occurred in January of 1700 and that such seismic events occur every 200 to 1,000 years, with an average return period of 500 years (CREW 2013).

A Cascadia Subduction Zone earthquake and tsunami event is considered one of the most severe potential disasters the state is likely to experience. Modeling of a Cascadia event estimates immediate deaths from earthquake shaking and tsunami inundation could exceed 10,000 across multiple states; direct economic losses could be upwards of $80 billion, depressing the economy of the region for decades (CREW 2013; FEMA 2015). This event dwarfs other nationally significant disasters, such as the 1994 Northridge earthquake in California, which caused $59.8 billion in direct losses (2006 dollars) and only 57 deaths. It is more comparable to the levee-topping Hurricane Katrina of

2005, which caused over $81 billion in property damages and killed over 1,500 people. However, the rapid-onset of the ensuing tsunami in the Northwest is expected to cause a dramatically higher number of casualties than in Hurricane Katrina (GAO 2007).

Loss estimation modeling of a Cascadia Subduction Zone earthquake has been extensive and is the basis for regional and national emergency planning (FEMA 2015). Such modeling estimates shaking to last on the order of several minutes, producing severe-intensity shaking along the coasts, with strong shaking within the populous Puget Sound Basin.[5] Of particular concern is the tsunamigenic potential of these earthquakes. A Cascadia Subduction Zone earthquake is likely to shift the seafloor upward abruptly, causing a tsunami wave to generate and close-in rapidly upon the Washington, Oregon, and British Columbian coastlines. The tsunami waves are expected to rise as high as nine to twelve meters by the time they hit shore; for the Pacific Coast in Washington state, that timing might be as little as 20 minutes after the earthquake and would be insufficient for thousands of residents to evacuate.[6] The earthquake and tsunami are expected to trigger further landslides and ground failure throughout the region, especially in mountain passes and in the steep mountain valleys of the Cascade and Olympic mountain ranges.

Population impacts

A daytime subduction zone earthquake is expected to injure over 20,000 people in Washington and Oregon, with further injuries expected once the tsunami arrives and aftershocks, fires, hazardous materials releases, and contaminated water supplies take their toll.[7] Altogether, medical facilities might experience a surge of as many as 30,000 injured survivors, many of whom would arrive at hospitals that were too damaged to administer care from within the facility. Injury, especially combined with exposure to inclement weather, would leave many residents weakened and less capable of physically adjusting to a period of food insecurity (FEMA 2015; Woods and Schmidtlein 2013; Woods and Soulard 2008).

Modeling of a Cascadia Subduction Zone earthquake and tsunami event suggests poor access to clean water as well as to usual and customary foods. It is quite likely that the Federal Emergency Management Agency would not deliver emergency food rations for at least a week, based upon emergency response in similar disasters discussed earlier. First served would be those along the populous I-5 corridor. Out-migration from coastal areas to urban population centers is quite possible, save for tribal and rural communities with strong cultural connections to their homes. As given in the subduction zone earthquake scenario, water would be severely compromised and food availability would be marginal. Decades ago, households may have relied on partially-cooked canned goods, as well as dry, packaged items in pantry stores. The changes in household food consumption patterns noted above make planning assumptions that most households would be able to be food self-sufficient for a week or more after a rapid-onset disaster increasingly inappropriate. Widespread food insecurity is a more likely reality. Although healthy

[5] The loss estimates discussed are based upon 90th percentile damage estimates — indicating a damage state with only a 10 percent estimated change of being exceeded. This credible worst-case scenario is used for regional planning purposes in Washington and Oregon and helps to account for secondary impacts of landslides and aftershocks not well modeled in the loss estimation program.

[6] Tsunami wave heights are expected to be much smaller and less damaging by the time they reach Seattle, Tacoma, and other urban centers within the Salish Sea; Vancouver Island will refract much of the wave energy back out into the Pacific Ocean basin.

[7] Loss modeling of injuries is based upon the time of day. Earthquake events that occur in the middle of the night tend to cause the least injury and death in the United States, as most people are in the relative safety of their homes. Events that occur during the daytime, especially during commuting hours, are expected to increase injuries by an order of magnitude or more.

adults may be less affected by a week or more of food insecurity,[8] vulnerable groups (children under five, pregnant and nursing women, malnourished and diseased, the elderly) may be especially vulnerable (WHO 2000).

Infrastructure impacts and implications for food distribution

Seismic shaking and ground failure are expected to severely degrade the region's transportation systems. Over a quarter of Washington state's road infrastructure in the I-5 corridor is expected to suffer medium to high damage. This level of damage means, at a minimum, large cracks and ground settlement; at the extreme, it means failure of pavement structure and subsurface materials. The situation is much bleaker west of I-5 along the Pacific Coast. There, three quarters of the roads are predicted to have high damage. Sections of U.S. Highway 101 along the coast are expected to suffer even higher levels of damage from the tsunami and be virtually unusable. Land routes to coastal communities could be cut off for weeks and months, shunting the bulk of emergency response to marine and air routes (FEMA 2015). Highly damaged roadways may require major repairs before vehicular traffic can resume; many damaged roadways may wait for months before road crews can make repairs.

Bridge damage would contribute to transportation degradation. For nearly 25 years, the state's Department of Transportation has invested in a $166 million bridge retrofitting program, which has shored up nearly 300 bridges. Twice as many still require work. However, the program has prioritized bridges in the dense, urban core of the Puget Sound from Olympia to Everett to create a transportation 'life-line' for transporting emergency supplies after a major earthquake. This work, while incomplete, is significant. In the populous Puget Sound Basin, irreparable bridge damage and collapse, is expected at only about 10 percent of bridges and highway overpasses. Another quarter are predicted to be moderately damaged; they are likely to require engineering assessments before being opened for vehicular transportation, and some may no longer be able to support heavy loads. Along the coast, however, an estimated 50 percent of bridges are expected to have high damage, meaning bridges could collapse or be in danger of collapse and impassable (FEMA 2015).

Of particular concern in Washington state is the lack of redundancy in its network of highways. With dense road networks in the I-5 corridor, alternative routes on surface streets may make local transportation connectivity — albeit with delays, longer routes, and restrictions on vehicle size —a possibility within several days of the event, even with some heavily-damaged bridges. However, many rural and coastal communities rely upon small highways that have few, if any, alternative routes. Also problematic is long-distance transportation connectivity between Western Washington and other areas. The state's mountainous geography has limited east-west transportation corridors to a handful of alternatives. A single major highway, Interstate 90, crosses the Cascade Mountains; six smaller east-west highways also cross but all would be compromised in the immediate aftermath. One pass routinely closes the entire winter and the others pass through steep valleys and ravines and would experience landslides that choke off transportation access for both roadway and railway. A single north-south highway transportation corridor, Interstate 5, connects all major population centers on the west side of the state and would be damaged. Loss estimations indicate that up to two-thirds of the interstate may suffer damages that seriously degrade or prevent transportation movement.

Significant and prolonged degradation of the state's road transportation infrastructure would be coupled with serious impacts to port facilities. Ports are typically built on unstable marine sediments and infill and are highly exposed to tsunami impacts. All state ports are expected to experience moderate to heavy damage. Piles supporting seawalls

[8] Existing observations of survivors of famine and concentration camps and other extreme events indicate that most healthy adults can survive without food for approximately 30 days and without water for several days (Packer 2002). While it is comforting to think that few healthy adults would die from lack of food access in a Cascadia event, the aim of community disaster planning around food security should be ensuring survivors do not experience a dramatic increase in food insecurity during the immediate aftermath and recovery from these events.

may be broken or damaged; cranes used for moving cargo between vessels and trucks or rail may topple. In highly damaged ports, the majority in the state, cranes may totally derail and warehouses may experience significant damage. The earthquake impacts are expected to result in partial or full failure of the docking facilities, rendering many residents virtually stranded (FEMA 2015). Such impact is significant; ferries, connecting the state's many island communities, carry approximately 22 million people and 10 million vehicles annually, and a disruption in service would be widely felt.

Transportation degradation — affecting road, rail, and port links — likely would impair replenishment of retail foods and other goods, especially for rural and coastal communities. However, it would likely also slow restoration of the electricity and natural gas that households typically use for cooking food. Vast swaths of western Washington would be expected to fall into complete blackout minutes after the earthquake as substation and distribution circuits fail and relays shut down power due to system overload. Repairs to transmission lines may take weeks to months; line crews would have to wait for emergency transportation crews to clear roads and shore up bridges and overpasses on critical routes before they could reach damaged substations and downed lines. Similarly, natural gas repair would require a minimally functional transportation system. Households may have their gas lines shut off for days in the Puget Sound area and for weeks to months in harder hit coastal communities (FEMA 2015).

Damage to drinking water and wastewater systems may also have immediate impact on survivors' food security. A majority of potable water systems in western Washington are estimated as likely to experience major damage, including complete failure of pipelines, extensive damage to treatment plants, and damage and collapse of water towers. Contamination from damaged wastewater pipes is expected throughout the region. Piped potable water may be unavailable for weeks to months, especially along the coast (FEMA 2015). As in the Christchurch earthquake, urban households may initially have to boil contaminated tap and surface water, which would require a fuel source, during the first weeks of the disaster. They then would likely need to rely upon potable water trucked to distribution sites for months thereafter. Rural households relying upon well water may retain access to potable water if they have manual pumping capacity or a generator and sufficient fuel. However, unpredictable shifts in water tables are common after earthquakes; some wells may go dry while new springs may emerge elsewhere.

Here, we examine options for food security during the immediate emergency period in the days and weeks following a Cascadia Subduction Zone earthquake and tsunami. For the purposes of our discussion, water is assumed to be available, though likely contaminated. Loss of, or only sporadic access to, electricity and natural gas also is assumed. Transportation networks are assumed to be highly degraded, with bridges and overpasses damaged and landslides and debris blocking many roads. In dense, urban environments, a patchwork of surface and arterial streets may quickly be cleared, allowing for repair and partial restoration of power and resumption of delivery to many food outlets after the first week. In rural areas and along the coasts, transportation repairs, and with it repairs to power distribution systems, may take several weeks. These assumptions are in line with current state emergency planning for a Cascadia Subduction Zone earthquake and tsunami. However, damages and restoration times used for planning are based upon complex probabilistic calculations; actual damages and restoration may vary substantially in an actual event.

Immediate food access

Food access during emergencies serves two functions: it prevents individuals from declining into an emergency health situation and reduces the likelihood of mass migration out of the region. Given that approximately 20 percent of Washington households rely on emergency food assistance programs each year, with very little safety cushion for emergencies, the potential for health impacts is significant. Further, mass migrations of individuals to a new area can compound a disaster by placing new burdens on an area that does not have the infrastructure to handle the demands for increased capacity. Mass out-migration also can have long-lasting effects on the economic recovery and evolving culture of a region (Hobor 2015). According to census data, nearly half of the King County population over the age

of 25 has a bachelor's degree or higher, as well as competitive professional skills; out-migration of educated, urban households from the Seattle metropolitan area also is a real possibility.

With degraded transportation and energy systems expected in a regional subduction zone earthquake, regional food availability and an organized storage and distribution method take on a critical new importance. Food that can be immediately accessed in the area may be more accessible than food sourced from outside western Washington, which would be reliant on badly damaged, long-distance transportation routes across the Cascade mountains, along the I-5 corridor, or through coastal seaports. In terms of food availability, existing food outlets (undamaged packaged food on the shelves of retail outlets following the earthquake) and sources (shelf-stable food in home pantries) could help supplement the slow trickle of emergency relief supplies brought in by air or on a patchwork of less damaged roadways. Certainly, the availability of high-volume fruits and vegetables in retail and food bank stocks would be uncertain.

Estimation of population need in a Cascadia Subduction Zone Event

Meeting basic nutritional needs would require the provision of about 1700 kcal of food per day, at least 20 percent or more of which would ideally be as fat (Seneff et al. 2011). The degree of food deficiency in the immediate aftermath of a subduction zone event depends upon food availability — household food stores, retail and food bank stock — the size of the vulnerable population, and the length of time before emergency relief supplies begin to reach affected communities. Such calculations, if they have been done at all, are not publically available. With depleting food sources, short-term and immediate food deficits would likely be in cereals (rice, cornmeal, or flours) and a concentrated source of fats or oils.

As an initial estimate, we can assume retail food stores and food banks have three days of stock but that emergency relief supplies may take an additional five to ten days to reach most affected populations. This leaves a potential two- to seven-day gap where households would need to be self-sufficient or have access to other food sources, such as regional food sources discussed later in this article.

For households that typically have stocked pantries, food security would seem to pose little problem. Most well stocked homes would be able to use existing stocks during the gap before emergency relief supplies arrived or retail restocking resumed. However, even well stocked households would find some food at home unusable after a large earthquake. Without electricity or backup generators, food in refrigerators and freezers would begin to spoil, depending on outside temperatures and how well the units were self-contained or airtight. Preserved food in glass jars might break and be unusable. Thus, many of these households could retain some undamaged, non-perishable food stores and might be able to maintain food security, albeit with a less varied diet.

In retail outlets, fresh produce, meat, dairy and other perishables would quickly spoil, unless stores had on-site backup generators and sufficient fuel. Broken glass would also likely litter the floors and force stores to close temporarily. When the remaining retail food stocks are depleted and initially fail to be replenished, households that rely upon frequent purchases of food would rapidly become food insecure after food stores at home are consumed. Food security would be even more threatened in households that rely upon visits to local food banks each week.

We estimate that western Washington residents would need on the order of four billion kcal in the gap between the depletion of home and retail food sources and the replenishment of retail or arrival of emergency response supplies five or more days later. As shown in Table 3, this estimate is based upon assessments of two vulnerable populations. The first is households heavily reliant upon the availability of retail food during a disruption. We assume that of the six million residents of western Washington, a million of these could fall into this category. We assume these individuals would have some access to food — either through limited household stores or purchase from un-depleted retail outlets — for three days, but would then dip into food insecurity for two days before emergency food

relief and restocking is widely available. Such populations may need access to roughly three billion kcal of food. The second vulnerable population would be individuals in households that were already struggling with food security when the earthquake strikes. Over 160,000 individuals visiting food banks each week in Washington state (Robinson 2014); we assume 80 percent of these individuals are in western Washington, equivalent to 128,000 individuals. We further assume these individuals have little access to any food after the first two days, leaving five or more days of food insecurity, amounting to one billion kcal. Together, these two vulnerable populations would need roughly four billion kcal of food for the region's surviving population to remain food secure in the immediate aftermath of the earthquake. Such calculations, while intended to provide a basic, first-order estimate of need, admittedly gloss over crucial aspects of geographic and temporal heterogeneity. Low-income households and physically vulnerable individuals in rural areas may be most at risk due to sluggish emergency transportation repairs in especially remote areas. We now turn to whether and how food, produced and stored regionally, could reduce food insecurity in this scenario.

Table 3. Potential caloric needs of western Washington population in the immediate aftermath of a subduction zone earthquake (author estimates)

	Population	Kcal	Days of food Insecurity	Total kcal
Heavily reliant upon retail food availability	1,000,000	1700	2	3 billion
Food Insecure at time of event	128,000	1700	5	1 billion
				4 billion kcal

Regional food for increased resilience

To identify regional food production, processing, storage, and distribution resources throughout the impacted areas, it is important to first look at data for Washington state as a whole. In 2012, Washington had approximately 40,000 farms with a total agricultural production of about $10 billion in sales (National Agriculture Statistics Service 2012). The top five non-meat food commodities are apples, wheat, milk, potatoes, and cherries; together they comprise over 50 percent of the state's agricultural output. Washington also is a major producer of hops, stone fruits, farm forest products, aquaculture fish, shellfish, onions, and mint oils. Such output represents a diversity of commodities. However, with much of supply going to the export market, production of these commodities contributes little to Washington state food security.[9] In fact, according to the USDA, Washington is the third largest exporter of agricultural commodities in the nation, exporting $9.3 billion in 2007 and $16.5 billion in 2012 (National Agriculture Statistics Service 2013).

Despite Washington's emphasis on globalized trade and sizable farm exports, 14 percent of farms in 2007 sold *directly* to consumers, restaurants, or retailers in Washington state, according to self-reported farm data collected by

[9] A statewide assessment (Green and Cornell 2015) suggests high quantities of food are produced and stored in eastern Washington. Stone fruits, grains, and potatoes produced on the Columbia Plateau and wheat and legumes in the Palouse could contribute to Washington state food security. However, transporting such food items to affected communities in western Washington would be hampered by bridge damage and landslides in mountain passes and along the I-5 corridor.

the U.S. Department of Agriculture's National Agricultural Statistics (Washington State Dept. of Health 2012). In contrast, according to the USDA's Economic Research Service, across the nation direct sales accounted for only 0.4 percent of total agricultural sales in 2007, up from 0.3 percent in 1997. It is not surprising then that Washington ranks seventh in the nation for having the largest direct farm sales, as well as second nationally for population-size-to-dollars-spent on direct farm sales. Perhaps the economic and social infrastructure is well in place for recognizing regional produce as a critical food source, contributing to resilience. In Washington, established and respected programs such as farm-to-school, farm-to-hospital and other institutions, community-supported agricultural share purchases, and farmer markets provide important regional outlets for farm produce.

Nevertheless, food system resilience is in jeopardy. Households rely heavily on imports, much of what they eat is pre-prepared, and many rely, at times, on federal provisions through food pantries. This suggests that an interruption to transportation and energy systems could represent a substantial disturbance in usual and customary eating.

Undoubtedly, continued household storage of non-perishable food and water for use in emergencies is essential, especially food that could be readily eaten without cooking and that is unlikely to be damaged during the shaking. Examples include packaged breakfast cereals and canned protein sources (meats, beans). Unfortunately, given the trend towards increased reliance upon out-of-home food expenditures and the significant number of households already relying upon food pantries, this solution alone is unworkable. The question we ask is, could regional food suppliers help provide the population with a caloric intake of 1700 kcal per person daily for several days to weeks immediately following a rapid-onset, regional disaster? The answer rests with both regional food production and distribution prospects, as well as with potable water and adequate fuels available for home preparation and cooking.

No centralized database or agency tracks regional food production and storage, especially for use in regional disasters. For example, does the dairy in Whatcom County or the green peas in Snohomish or the potatoes in Skagit put each of these counties on the map as a potential source of calories, either during an incident or throughout the year? It seems unlikely, given both insufficient quantities and seasonal-only availability. More likely sites would be wholesale food processors and storage, such as seafood processors in Anacortes and Bellingham and those more centrally located along the I-5 corridor in Skagit County.[10] These processing/storage facilities take in significant amounts of food product from regional producers. For example, five seafood processing and cold storage facilities in Skagit County hold an estimated 100 million kcal combined at any given time (Green and Cornell 2014). These cold storage sites may be an immediate source of food, independent of season, though they make barely a dent in the potential need. Thus, we must turn to sources that are more novel.

From a regional foodshed perspective, one source of protein-rich fat is hazelnuts, which also are high in fiber, monounsaturated fatty acids, tocopherols (Vitamin E compounds), squalene (important in the synthesis of sterols), and phytosterols.[11] Small grains, which include wheats, perennial wheat, and other perennial cereal crops, are another source of carbohydrates and protein (Maguire et al. 2004). A single cup of whole hazelnuts with a cup of whole, dry grains can provide almost 800 high-quality kcal. Both hazelnuts and grains are easily stored. Focusing on wheat certainly makes sense as it is a high-demand cereal crop, and increasingly so in the Pacific Northwest as new winter wheat cultivars adapted to Washington state are being developed (Hayes et al. 2012; Hills et al. 2013; Jakumar et al. 2012; Jones 2012). The question germane to this article is whether these cultivars could be grown on such a scale as to supply short-term, immediate food assistance markets.

[10] Several seafood processors and cold storage facilities along the Pacific Coast are within the tsunami inundation zone.

[11] Note that virtually all the hazelnuts in the U.S. are produced in Oregon as reported by the Agricultural Marketing Resource Center and USDA ERS statistics on fruit and tree nuts, although Washington state is gaining ground. Hazelnuts in Oregon would necessarily better support affected communities in that state.

The answer to whether western Washington farms can produce high volumes of staples regionally is a qualified yes, especially for perennial grains (Hills et al., 2013; Murphy et al., 2009; Selfa and Qazi, 2005). Annually, 30-50,000 tons of perennial grains are grown each year in Skagit and Whatcom counties alone; at roughly 1500 kcal per pound, this represents many times the need estimated above. Dry beans, also, are becoming much more prominent (Washington State University Research and Extension Center, personal communication, May 28, 2015). It would seem that, in the face of a regional disaster, it is possible for regional farms to produce sufficient grains and beans annually to provide immediate, post-disaster food security for western Washington households. Yet, to be feasible as an emergency food source, current and expanded production would need to be embedded in a regional system of production and storage.

From Vancouver Island, British Columbia to Mount Vernon, Washington, to Asheville, North Carolina small-scale grain growing is a real possibility. What would it be like to imagine a staple production system that reflected, "the true cost of production . . . [with] independent production, processing, and distribution networks" (Hill et al. 2011, p33). The authors refer to the discovery of grain-growing potential as reclaiming a grain heritage. It also is establishing a diversity of markets since small-grain crops may be grown in rotation with different higher-value crops.

Hill et al. emphatically state that a locally integrated small-grain system is regionally possible, thus creating shorter supply chains. Infrastructure supporting this was available in the past, less so now that urban planning emphasis is devoted to metropolis needs. Such considerations, though, go beyond imagining niche markets for regional malted barley and a locally produced pretzel. They also speak to the very real possibility of a strong small-grains production system, producing for immediate use as well as for long-term storage up to, say, two years (Higginbotham et al. 2011).

Prospects and challenges of regional storage, distribution, and usage

While regional staple production is feasible, storage and delivery are germane to the discussion of making these sources a real possibility in a disaster. According to the Washington State Department of Agriculture, while there are no formal agreements or contracts to ensure continuity of locally-sourced, fresh produce deliveries during a regional disaster or major supply chain disruption, plans are underway to change the situation. Big, nonprofit storage facilities are also growing throughout the state, and so is their storage capacity. This is mostly for commodity foods, i.e., foods distributed under federal assistance such as The Emergency Food Assistance Program. However, assuming storage capacity increases, nuts, grains, and beans, some derived from farmer-distributor direct contracts, could be stored in turn. Big storage projects already underway in western Washington include Thurston County Food Bank and Northwest Harvest and Food Lifeline in King County (Washington State Dept. of Agriculture, personal communication, May 28, 2015). These organizations already work with local farms and manufacturers to identify food waste situations, arranging to have nutritious food — that would otherwise be discarded — be redirected to food banks.[12]

The next challenge is distribution. While regionally stored foods would likely have shorter travel distances and less disruption than food from outside the region, even distribution of regionally stored foods would have to contend with a damaged transportation system. According to Washington State Department of Agriculture Emergency

[12] Due to prohibitions on use of federal monies, food stored in contracted warehouses, such as food banks, may not be used for disaster-related assistance until authorized. Note that the Washington State Emergency Operations Center would be responsible for deciding and communicating with the county food banks before contracted warehouses could modify their standard food distribution protocol.

Preparedness group representatives, although delivery time would be increased, there are both land and sea routes to deliver products within a few days to most of the state. One notable exception would be moving supplies to counties south of Puget Sound, where I-5 disruptions would seriously impact delivery (Washington State Dept. of Agriculture, personal communication, May 28, 2015). If other disasters are any indication, a surge of volunteers and local residents would use motorbikes or four-wheel vehicles, physically carrying items around road blockages to bring the first emergency supplies of food, bottled water, and medicine to surviving rural and coastal communities. Strategies that reduce the distance between food supply and survivor, such as strategically placed regional food storage, may decrease the time and difficulty these volunteers have in reaching survivors, especially if such regional food storage and volunteer planning are integrated prior to an event.[13] Such strategic storage would be especially beneficial for rural and coastal communities west of I-5.

We believe a proactive approach and emergency plan could be developed to bolster a regional food strategy to help minimize impacts of a supply chain disruption. Such an approach could include entering into agreements or developing emergency contact frameworks for warehouses capable of providing significant calories and flexible enough to redirect their operations during a disaster as an additional or temporary market. Emergency managers, local jurisdictions, and food banks could also develop interagency agreements with each other and larger agencies such as the Washington State Department of Agriculture and the U.S. Department of Agriculture, a major supplier to food banks. These interagency agreements could increase the adaptive capacity of emergency food distribution systems during a regional disruption. Separately, direct agreements between local growers and food banks could improve access to fresh produce, supplementing grain and nut storage.

To make these regional sources functionally available, households would need to be advised how to cook or process grains, legumes, and nuts. Many households have become unaccustomed to producing edible and nutritious meals from standard ingredients, let alone doing so in a disaster. Using unprocessed local grains and nuts as an immediate food source would be predicated on households boiling water and preparing these products on open fire or portable, gas cook stoves and grills. For many rural and suburban households in western Washington, this requirement may be within the realm of possibility; for others, such as urban apartment dwellers, the requirement may be more problematic. Neighborhood-level 'soup' kitchens may provide immediate and ongoing relief for households without access to cooking fuel. Food banks may be an ideal location for large-scale cooking and distribution of regional foods. In all cases, increasing palpability through adding salts, sugars, and spices may also need to be addressed.

Conclusion

Multiple and diverse food sources and distribution channels are important for building resilient regional food systems able to support communities in the immediate aftermath of a rapid-onset natural hazard. For Washington state, the current trend of manufactured, ready-to-eat foods, and increasing food imports places a profound importance on the transportation infrastructure. Any scenario that degrades transportation networks would do more than reduce food on grocery store shelves. Degraded transportation also would drastically hamper emergency food assistance programs that rely overwhelmingly on USDA-supplied and other manufactured food originating outside the region. While a rapid-onset hazard like a Cascadia Subduction Zone earthquake and tsunami is unlikely to produce severe and prolonged food insecurity for the vast majority of the population, many households in the region

[13] Also important to consider in terms of food delivery is the concept of 'emergency relief chains,' i.e., links between customer services and non-profit, non-routinized supply chains. Examples attest to the ability of regional food networks to provide scheduled food assistance, such as school lunch programs (Sanger and Zenz 2004). In terms of short-term relief, infrastructure and relief program organization remain challenges. Better understanding of what motivates such 'chains' in a humanitarian context is needed (Oloruntoba and Gray 2009). Further, a more empathetic distribution system would also prioritize palatability in provision of usual and customary foods.

could experience a food 'gap,' running out of food before emergency relief efforts and restocking have fully reached them. Such a gap is unlikely to cause widespread fatalities. Exposure to wet and cold weather are more troublesome in that regard. However, even short-term food insecurity can increase anxiety, lead to risky behavior, and prompt widespread out-migration that can overwhelm receiving communities or have long-term impacts on regional recovery.

Ensuring food security even during the immediate aftermath of a major natural hazard should be addressed by a myriad of overlapping mitigation and preparedness actions. At the state level, continued funding of transportation retrofitting, especially bridges and overpasses along critical routes, would strengthen the regional transportation network. A strengthened network would require less repairs and allow conventional retail and emergency food to reach impacted communities more quickly. It would also speed repairs of electrical, water, and natural gas networks. At the household level, emergency managers and community organizations need to continue to encourage households to store (and secure) non-perishable, ready-to-eat food and sufficient water, reminding them that grocery store stock and the out-of-home eating on which they are increasingly reliant may be intermittent or unavailable for days, even weeks, after a Cascadia event. Households should be encouraged to build slowly stockpiles of food and water, enough for several days, even up to several weeks. These approaches are well known, but we argue that they should be enhanced by regional actions.

Regional, diverse agriculture is part of the answer to post-disaster food security. Planning with dual aims to bolster food system resilience and farm economies is particularly desirable. Concerted regional storage and distribution efforts may help fill the 'gap' between the end of household and retail stocks and the arrival of robust emergency relief. Yet, in order for local foods to provide significant calories in a crisis, we need to have in place a robust, and regionally appropriate, food resilience strategy. This needs to be a strategy that re-balances our dependence on foods supplied through imports and long supply chains, as well as increasing our reliance on regional food sources. A regional food resilience strategy could be strengthened by diversification in agricultural production, especially of perennial grains. It nevertheless would need to be augmented by a regional storage and adaptive distribution strategy. It could be advanced by greater public awareness and policy in support of healthier diets. Such diets could include, in addition to fruits and vegetables, whole grains, dry beans, and tree nuts.

Currently, with the limited number of small- and medium-sized farms surrounding urban areas, regional farms would likely be unable to support a long-term disaster — were widespread food insecurity to occur over weeks and months, emergency food planning would need to consider diverse dietary needs and nutrition requirements beyond simply providing adequate calories. However, production of storable crops such as grains, beans, and nuts could contribute to food resilience strategies for both disasters that caused short- and longer-term gaps between event and relief. Already, researchers at the Washington State University Mount Vernon Research Station are working to promote widespread regional production of small grains and dry beans.

There is untapped potential, however, for regional farms to contribute a significant amount in the interests of food system resilience if the appropriate institutional frameworks, price incentives, and infrastructure maintenance schedules were to be implemented. Under current models of emergency food assistance, independent non-profit organizations are critical in developing a new framework since they are more flexible in arranging with growers, who may be hesitant to work with government agencies. These independent non-profits generally have fewer policies restricting cooperative arrangements, as well as less bureaucracy, so they can achieve more timely results.

Much has been written about the "local" and "regional" in food, but little in terms of food interruptions and increased food system resilience. Could extreme situations suggest the need for greater reliance in more ordinary times? The results of this research indicate that the potential for a greater contribution of regional farm production into mainstream supply, especially in the interests of food system resilience, could be substantial. This is the case especially in a state like Washington, where direct sales, indicative of support for regional production, currently

constitute a large percentage of the total food market. Greater reliance on local sources of food, in and outside disasters, may increase regional food resilience.

Compliance with Ethical Standards

All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

References

Belyakov A (2015) From Chernobyl to Fukushima: an interdisiciplinary framework for managing and communicating food security risks after nuclear plant accidents. J Environ Stud Sci 5:404-417

Berardi G, Green R, Hammond B (2011) Stability, sustainability, and catastrophe: applying resilience thinking to U.S. agriculture. Hum Ecol Rev 18:115-25

Clancy K, Ruhf K (2010) Is local enough? some arguments for regional food systems. Choices Magazine, Agricultural and Applied Economics Association. http://www.choicesmagazine.org/magazine/article.php?article=114 Accessed 20 May 2015

Colman-Jensen A, Gregory C, Singh A (2014) Household food security in the United States in 2013. Economic Research Report No. 173, United States Department of Agriculture. http://www.ers.usda.gov/publications/err-economic-research-report/err173.aspx. Accessed 27 May 2015

CREW (2013) Cascadia subduction zone earthquakes: a magnitude 9.0 earthquake scenario. http://crew.org/sites/default/files/cascadia_subduction_scenario_2013.pdf Assessed 27 May 2015

Dilleya M, Boudreaub TE (2001) Coming to terms with vulnerability: a critique of the food security definition. Food Policy 26:229-247

Dubbeling, M (2013) Linking cities on urban agriculture and urban food systems. RUAF Foundation. http://www.ruaf.org/sites/default/files/Text%20city%20food%20systems.pdf Accessed 16 Jul 2014

ERS (2015) U.S. food imports. USDA Economic Research Services. http://www.ers.usda.gov/data-products/us-food-imports.aspx Accessed April 30 2015

Endres AB, Endres JM (2009) Homeland security planning: what victory gardens and Fidel Castro can teach us in preparing for food crises in the United States. Food Drug Law J 64:405-39

Engle PL Castle S, Menon P (1996) Child development: vulnerability and resilience. Food Consumption and Nutrition Division of the International Food Policy Research Institute, Discussion Paper No. 12. http://ageconsearch.umn.edu/bitstream/97309/2/child%20development.pdf Accessed May 30 2015

FEMA (2015) Cascadia rising: cascadia subduction zone earthquake and tsunami exercise scenario document. https://huxley.wwu.edu/sites/huxley.wwu.edu/files/media/Cascadia_Rising_low_1.pdf Assessed 13 April 2015

FDA (2013) Global engagement. http://www.fda.gov/downloads/AboutFDA/ReportsManualsForms/Reports/UCM298578.pdf Accessed 27 May 2014

Food logistics (2014) Key ports and carriers in the global food supply chain. http://www.foodlogistics.com/article/12002254/key-ports-carriers-in-the-global-food-supply-chain. Accessed 15 April 2015

Fraser ED, Legwegoh A, Krishna KC (2015) Food stocks and grain reserves: evaluating whether storing food creates resilient food systems. J of Environmental Studies and Sciences 5:445-458

GAO (2007) Natural hazard mitigation: various mitigation efforts exist, but federal efforts do not provide a comprehensive strategic framework. Report GAO-07-403. http://www.gao.gov/new.items/d07403.pdf Accessed 28 May 2015

Green R, Cornell J (2014) Regional market analysis of food security and regional resilience: whole community preparedness through local food production and distribution in Washington state. https://huxley.wwu.edu/sites/huxley.wwu.edu/files/media/2014_RI_REPORT_FEMA_Food_Security_0.pdf Accessed 1 September 2015

Guthrie J, Lin B, Okrent A, Volpe, R (2013) Americans' food choices at home and away: how do they compare with recommendations? USDA Economic Research Service. http://ers.usda.gov/amber-waves/2013-february/americans-food-choices-at-home-and-away.aspx#.U76pH1a4mlJ Accessed 10 June 2014

Hammond B, Berardi G, Green, R (2013) Resilience in agriculture: small- and medium-sized farms in northwest Washington state. J Sustainable Agric 37:316-339

Hayes RC, Newell MT, DeHaan LR, Murphy K, Crane S, Norton MR, Wade LJ, Newberry M, Fahim M, Jones SS, Cox TS, Larkin PJ (2012) Perennial cereal crops: an initial evaluation of wheat derivatives. Field Crops Res 133:68-89

Higginbotham RW, Jones SS, Carter AH (2011) Adaptability of wheat cultivars to a late-planted no-till fallow production system. Sustainability 3: 1224-1233

Hills K, Corbin A, Jones S (2011) Rebuilding the grain chain: stories from the coastal Pacific Northwest. Rural Connections Sept: 32-36. http://wrdc.usu.edu/files/publications/publication/pub__9828019.pdf. Accessed 25 Sept 2015

Hills KM, Goldberger JM, Jones SS (2013) Commercial bakers and the relocalization of wheat in western Washington state. Agr Hum Values 30:365-378

Hobor G (2015) New Orleans' remarkably (un)predictable recovery: developing a theory of urban resilience. Am Behav Sci 59:1214-1230

Hori M, Iwamoto K (2014) The run on daily foods and goods after the 2011 Tohoku earthquake: a fact finding analysis based on homescan data. Japanese Political Economy 40:69-113

HVRI (2013) U.S. Hazard losses 1960-2013. http://hvri.geog.sc.edu/SHELDUS/index.cfm?page=products#State Accessed June 11, 2015

IRS (2014) Food industry overview. Report LMSB-04-0207-018. http://www.irs.gov/Businesses/Food-Industry-Overview. Accessed 15 Jul 2014

Irwin JD, Ng VK, Rush TJ, Nguyen C, He M (2007) Can food banks sustain nutrient requirements? a case study of southwestern Ontario. C J Public Health 98:17-20

Jakumar N, Snapp S, Murphy K, Jones SS (20 12) Agronomic assessment of perennial wheat and perennial rye as cereal crops. Agron J 104:1716-1726

Jones SS (2012) Kicking the commodity habit: on being grown out of place. Gastronomica 12:74-77

Kloppenburg J, Hendrickson, J, Stevenson, GW (1996) Coming in to the foodshed. Agr Hum Values 13:33-42

Low S, Vogel S (2011) Direct and intermediated marketing of local foods in the United States. USDA Economic Research Report No. 128. http://www.ers.usda.gov/media/138324/err128_2_.pdf Accessed 23 May 2015

Maguire LS, O'Sullivan SM, Galvin K, O'Connor TP, O'Brien NM (2004) Fatty acid profile, tocopherol, squalene and phytosterol content of walnuts, almonds, peanuts, hazelnuts and the macadamia nut. Int J Food Sci Nutr 55:171-178

Marchione T (2002) Foods provided through U.S. government emergency food aid programs: policies and customs governing their formulation, selection and distribution. J Nutr 132: 2104S–2111S

Martinez S, Hand M, Da Pra M, Pollack S, Ralston K, Smith T, Vogel S, Clark S, Lohr L, Low S, Newman C (2010) Local food systems: concepts, impacts, and issues. USDA Economic Research Service Report No. 97. http://www.ers.usda.gov/media/122868/err97_1_.pdf Accessed 30 Jul 2014

McColl G, Burkle F (2012) The new normal: twelve months of resiliency and recovery in Christchurch. Disaster Medicine and Public Health Preparedness 6: 33-43

Moises C (2007) Brown bag blues: New Orleans residents can't buy fresh produce. New Orleans CityBusiness 1.

Morange M (2013) Food bank fresh aids Whatcom county farmers, ensures produce for clients. http://agr.wa.gov/FoodProg/docs/FoodBankFresh.pdf Accessed 20 Mar 2015

Mundorf AR, Willits-Smith A, Rose D (2015) 10 years later: changes in food access disparities in New Orleans since hurricane Katrina. J Urban Health 92:605-610

Murphy K, Hoagland A, Reeves P, Baik B, Jones S (2009) Nutritional and quality characteristics expressed in 31 perennial wheat breeding lines. Renew Agr Food Syst 24: 285-292

National Agricultural Statistics Service (2012) The pride of Washington state. United States Department of Agriculture. http://www.nass.usda.gov/Statistics_by_State/Washington/Publications/wabro.pdf Accessed 30 Jun 2014

National Agricultural Statistics Service (2013) Value of Washington's 2012 agricultural production surpasses previous year's record high. United States Department of Agriculture. http://agr.wa.gov/AgInWa/docs/2012WaAgValuesUSDAPressRelease.pdf Accessed 7 Jun 2014

Oliver-Smith A (1999) Peru's 500-year earthquake: vulnerability in historical context. In: Oliver-Smith A, Hoffman S (eds) The angry earth: disaster in anthropological perspective. Routledge, New York, pp 74-88

Oloruntoba R, Gray R (2009) Customer service in emergency relief chains. Int J Phys Distrib 39:486–505

Owen OE, Smalley KJ, D'Alessio DA, Mozzoli MA, Dawson EK (1998) Protein, fat, and carbohydrate requirements during starvation: anaplerosis and cataperosis. J Clin Nutr 68:12-34

Packer RK (2002) How long can the average person survive without water? Sci Am. http://www.scientificamerican.com/article/how-long-can-the-average Accessed 15 June 2014

Poppendieck J (1994) Dilemmas of emergency food: a guide for the perplexed. Agr Hum Values 11:69-76

Prowse M (2003) Towards a clearer understanding of 'vulnerability' in relation to chronic poverty. Chronic Poverty Research Centre Working Paper No. 24. http://www.chronicpoverty.org/uploads/publication_files/WP24_Prowse.pdf Accessed May 29 2015

Robinson K (2014) Annual report, Food Safety and Consumer Services, Washington State Department of Agriculture. http://agr.wa.gov/FoodProg/docs/FSCS_Annual_Report.pdf Accessed 1 Oct 2015

Rose A, Wing IS, Wei D, Avetisyan M (2012) Total regional economic losses from water supply disruptions to the Los Angeles county economy. Los Angeles County Economic Development Corporation. http://www.laedc.org/reports/WaterSupplyDisruptionStudy_November2012.pdf Accessed Sept 3 2015

Rotary First Harvest (2013) Washington state growers roundtable report - farmers first. http://firstharvest.org/harvest-against-hunger/grower-round-table/ Accessed 2 Jul 2014

Ruhf K (2015) Regionalism: a New England recipe for a resilient food system. J Environ Stud Sci. doi: 10.1007/s13412-015-0324-y

Sanger K, Zenz L (2004) Farm-to-cafeteria connections: market opportunities for small farms in Washington state. Washington State Department of Agriculture. http://agr.wa.gov/Marketing/SmallFarm/docs/102-FarmToCafeteriaConnections-Web.pdf Accessed 20 May 2015

Seneff S, Wainwright G, Mascitelli L (2011) Is the metabolic syndrome caused by a high fructose, and relatively low fat, low cholesterol diet? Arch Med Res 7:8-20. doi:10.5114/aoms.2011.20598

Selfa T, Qazi J (2005) Place, taste or face-to-face? Understanding producer-consumer networks in "local" food systems in Washington state. Agr Hum Values 22: 451-464

Starkey JL (1994) An evaluation of emergency food bags. J Can Diet Assoc 55:175–178

Steinberg T (2000) Acts of god: the unnatural history of natural disaster in America. Oxford University Press, New York

Taylor A (2012) Hurricane Sandy: the aftermath. The Atlantic http://www.theatlantic.com/photo/2012/11/hurricane-sandy-the-aftermath/100397/ Accessed 12 June 2015

Thomas JA, Mora K (2014) Community resilience, latent resources and resource scarcity after an earthquake: is society really three meals away from anarchy? Nat Hazards 74:477-490

U.S. Dept. of Commerce (2008) Food manufacturing. US DOC Industry Report NAICS 311. http://trade.gov/td/ocg/report08_processedfoods.pdf Accessed 8 Jun 2014

Washington State Dept of Health (2012) Report on Washington's food system response to executive order 10-02. With Washington State Department of Social and Health Services, the Superintendent of Public Instruction, and the Washington State Department of Agriculture. http://depts.washington.edu/uwcphn/work/php/Washington%27s_Food_System_Report_01_17_12.pdf Accessed 10 Mar 2015

Washington State University Extension Service (2012) Skagit county agricultural statistics. http://ext100.wsu.edu/skagit/agriculture Accessed 4 May 2014

WHO (2000) The management of nutrition in emergencies. http://whqlibdoc.who.int/publications/2000/9241545208.pdf Assessed 27 May 2015

Wisner B, Blaikie P, Cannon T, Davis I (2003) At risk: natural hazards, people's vulnerability and disasters. 2nd ed. Routeledge, New York

Wood N, Soulard C (2008) Variations in community exposure and sensitivity to tsunami hazards on the open-ocean and Strait of Juan de Fuca coasts of Washington. USGS Scientific Investigations Report 2008-5004

Wood N, and Schmidtlein M (2013) Community variations in population exposure to near-field tsunami hazards as a function of pedestrian travel time to safety. Nat Hazards 65:1603-1628

Weber CL, Matthews SH (2008) Food miles and the relative climate impacts of food choices in the United States. Environ Sci Technol 42:3508-2513